

Integration of Satellite-Derived HCHO/NO₂ Ratios for Regime-Aware Tropospheric Ozone Modeling in Delhi

1. Introduction to the Photochemical Landscape of Delhi

The National Capital Region (NCR) of Delhi, situated within the broader Indo-Gangetic Plain (IGP), represents one of the most chemically complex and heavily polluted atmospheric

environments globally. Tropospheric ozone (O_3) pollution in this region is dictated by a highly non-linear interaction between nitrogen oxides ($NO_x = NO + NO_2$) and volatile organic compounds (VOCs). These precursor emissions are heavily modulated by intense incident solar radiation, extreme seasonal variations in planetary boundary layer (PBL) dynamics, and massive aerosol optical depth (AOD) loadings¹.

For numerical modeling of ground-level ozone, capturing the explicit chemical mechanisms of

the VOC- HO_x - NO_x cycle is the standard, albeit computationally expensive, approach.

However, operational constraints frequently limit the availability of high-fidelity VOC emission inventories and real-time organic boundary conditions. When attempting to model ground-level ozone exclusively through the aerosol-photolysis pathway—utilizing a subset of the Master

Chemical Mechanism (MCM v3.3.1) restricted to clear-sky $J(NO_2)$ and $J(O^1D)$ photolysis rates attenuated by AOD and the solar zenith angle (SZA)—the model fundamentally simulates an inorganic photostationary state (PSS) rather than true net ozone production.

To bridge the substantial gap between a computationally efficient, photolysis-only chemical state and the non-linear realities of urban ozone chemistry, the ratio of formaldehyde to nitrogen

dioxide ($HCHO/NO_2$, hereafter referred to as the Formaldehyde-to-Nitrogen Dioxide Ratio or FNR) is frequently deployed as a satellite-derived proxy for ozone production sensitivity.

Because HCHO is a high-yield intermediate oxidation product of nearly all non-methane volatile organic compounds (NMVOCs) and has a relatively short atmospheric lifetime (roughly one to nine hours), its spatial distribution closely mirrors the presence of reactive organic precursors, making it an effective indicator of total VOC reactivity³. By mapping the FNR across a model

domain, one can theoretically identify the spatial distribution of NO_x -limited and VOC-limited (radical-limited) regimes.

This comprehensive report provides an exhaustive analysis of the deployment of FNR maps to inform an aerosol-photolysis constrained ozone model over Delhi. The analysis defines the specific threshold values required for Indian megacities (moving beyond outdated global

defaults), details the retrieval physics and biases of the Sentinel-5 Precursor (S5P) TROPOspheric Monitoring Instrument (TROPOMI) products, identifies pre-gridded cloud computing architectures for seamless data extraction, compares the statistical validity of daily dynamic tracking against seasonal climatologies, and delivers a concrete data processing recipe. Crucially, the report evaluates the foundational physical chemistry to provide an unvarnished assessment of the limitations inherent in applying an FNR regime label to a numerical model entirely devoid of explicit VOC kinetics.

2. The Physical Chemistry of Delhi's Ozone and the Photolysis Constraint

To understand how an FNR map interacts with an MCM v3.3.1 photolysis-only configuration, it is necessary to rigorously define the chemical mechanisms at play. The stated model framework restricts itself to the aerosol-photolysis pathway, completely lacking VOC measurements and explicit VOC degradation mechanisms.

2.1 The Leighton Relationship and the Photostationary State

In the troposphere, the only known pathway for the in-situ production of ozone is the photolysis of nitrogen dioxide (NO_2), which produces atomic oxygen that subsequently reacts with molecular oxygen. The primary reactions are:

1. $NO_2 + h\nu \rightarrow NO + O(^3P)$ (driven by the photolysis rate J_{NO_2})
2. $O(^3P) + O_2 + M \rightarrow O_3 + M$ (where M is a stabilizing third body)

However, the ozone generated is rapidly destroyed by the nitric oxide (NO) produced in the first step:

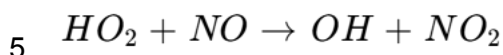
3. $O_3 + NO \rightarrow NO_2 + O_2$ (driven by the kinetic rate constant k_3)

In a purely inorganic atmosphere devoid of VOCs—which is precisely what a photolysis-only MCM configuration simulates—these three reactions form a rapid null cycle, known as the Leighton relationship or the photostationary state (PSS). The steady-state concentration of ozone is analytically given by the balance of these reactions:

$$[O_3]_{PSS} = \frac{J_{NO_2} [NO_2]}{k_3 [NO]}$$

In this closed chemical loop, there is zero net creation of ozone. To generate an accumulation of ground-level ozone that exceeds background levels, the atmospheric system requires an alternative chemical pathway to convert NO back into NO_2 without consuming an O_3 molecule. In the real atmosphere, this conversion is accomplished by hydroperoxy (HO_2) and organic peroxy (RO_2) radicals generated by the oxidation of VOCs:

4. $RO_2 + NO \rightarrow RO + NO_2$



2.2 The Purpose of the FNR Regime Label

The fundamental purpose of an FNR regime map is to dictate whether the net production of ozone via reactions 4 and 5 is limited by the availability of the radicals (a VOC-limited regime) or

by the availability of the NO_x molecules (a NO_x -limited regime)⁴.

In a VOC-limited regime, typically found in highly polluted urban cores like central Delhi during winter, massive emissions of NO_x from vehicular traffic and industrial combustion overwhelm the radical pool. The primary hydroxyl radical (OH) is scavenged by NO_2 via the termination reaction $NO_2 + OH \rightarrow HNO_3$ faster than it can oxidize VOCs⁴. Under these conditions, adding more NO_x actively suppresses ozone formation (the "titration"

effect), and ozone levels only rise if VOC emissions increase or NO_x emissions decrease. This exact phenomenon was widely observed in Delhi during the COVID-19 lockdowns of 2020; despite massive reductions in traffic NO_x , surface ozone concentrations actually increased in several urban sectors because the atmosphere was deeply VOC-limited⁹.

In a NO_x -limited regime, typically found in rural environments, downwind plumes, or during the summer monsoon when biogenic VOC emissions are highly abundant, the radical pool is large enough that every available NO molecule is rapidly oxidized to NO_2 . Ozone production in this regime scales almost linearly with the introduction of new NO_x emissions¹. Because the target modeling framework lacks the VOC chemistry required to compute the RO_2 and HO_2 radical budgets, the model cannot organically compute net ozone production regardless of the incoming solar radiation. The FNR regime map must therefore function as an external empirical constraint, informing the model of the ambient chemical state that it cannot simulate itself.

3. Spatiotemporal FNR Thresholds for Indian Megacities

3.1 The Evolution and Failure of Traditional Thresholds

The foundational premise of the FNR approach relies on substituting the radical loss ratio (L_{RO_x} / L_{NO_x}) with the observable tropospheric column ratio of HCHO to NO_2 . Early atmospheric modeling studies over the United States established traditional threshold values suggesting that an FNR less than 1.0 indicates a VOC-limited regime, values between 1.0 and 2.0 indicate a transitional regime, and values greater than 2.0 indicate a NO_x -limited regime⁴.

Applying these default US-based or generic Chinese thresholds to the Indo-Gangetic Plain and Delhi specifically results in severe misclassifications of the photochemical state. The traditional boundaries were derived from older chemical transport models (like CMAQ) tuned to North American emission profiles, moderate boundary layers, and lower background aerosol concentrations.

Recent observational and high-resolution modeling studies have fundamentally challenged the predictive power of these static thresholds. Analyses indicate that the relationship between FNR

and the radical loss ratio L_{ROx}/L_{NOx} is robustly linear only on a logarithmic scale⁴.

Furthermore, the transitioning ratios do not form a sharp, universal cutoff. A statistical evaluation across diverse regimes reveals that the transition zone strictly follows a Gaussian distribution with a mean of 1.8 and a standard deviation of 0.4. This statistical variance implies that the satellite-derived FNR inherently contains a 20% standard error resulting from the inability to

perfectly parameterize the full complexity of the RO_x - HO_x cycle solely through HCHO and NO_2 column measurements⁴.

Consequently, modern observational frameworks define the transitional regime boundaries much higher than traditional models. Alternative observation-based threshold methodologies,

such as those derived by connecting OMI HCHO/ NO_2 ratios to high ozone event probabilities calculated from in-situ ground monitors, suggest that VOC-limited conditions can persist up to

an FNR of 3.2, with the transition zone spanning 3.2 to 4.1, and NO_x -limited regimes occurring only above 4.1⁵. While this specific 3.2 threshold was derived utilizing US datasets, recent studies over Chinese megacities exhibit similar regional variability, with transition points ranging from 2.2 up to 3.7 depending on specific regional environmental conditions such as radiation intensity and surface temperature⁵.

3.2 Thresholds Specific to Delhi and the Indo-Gangetic Plain

Megacities in the Global South, characterized by extreme ground-level NO_x concentrations from dense traffic and unique VOC profiles stemming from mixed biomass burning, biogenic emissions, and unregulated industrial solvents, exhibit upwardly shifted FNR thresholds¹. In

Delhi, the extreme abundance of ground-level NO_x routinely suppresses the radical pool, driving the city center deeply into the VOC-limited regime during peak pollution hours, even when the bulk column FNR suggests a transitional state.

The photochemical regime is heavily modulated by the Indian monsoon cycle, driving extreme meteorological seasonality. Ground-based Multi-Axis Differential Optical Absorption Spectroscopy (MAX-DOAS) measurements in the IGP, specifically near Delhi and Mohali, demonstrate that the ozone production regime undergoes massive seasonal shifts that satellite thresholds must account for².

During the winter months (December to February), the planetary boundary layer over Delhi routinely collapses to a few hundred meters due to severe temperature inversions, trapping

traffic, heating emissions, and agricultural residue burning near the surface¹⁰. Despite significant local emissions of oxygenated VOCs from biomass burning, the reduction in incident solar radiation restricts the photolysis-driven generation of the primary hydroxyl radical (OH). The intense accumulation of NO_2 pushes the Delhi urban core deeply into a VOC-limited state. During this season, MAX-DOAS studies confirm that the FNR drops significantly, and the peak daytime ozone production regime becomes sensitive to both NO_x and VOCs, leaning heavily toward VOC-limitation in the early morning (07:00 to 09:00 LT) before shifting to a transitional state for the remainder of the day².

Conversely, during the summer (March to May) and the Indian Summer Monsoon (June to September), higher temperatures exponentially increase the emission of highly reactive biogenic VOCs (such as isoprene) from the surrounding vegetation across the IGP¹⁵. Concurrently, deep convection and elevated boundary layers (frequently exceeding 2,000 meters) efficiently ventilate urban NO_x , while heavy monsoonal rains selectively wash out soluble pollutants². During these periods, the production of HO_x radicals is maximized, and the FNR frequently exceeds the upper transition bounds, transitioning the Delhi region into a strongly NO_x -limited regime².

Based on the synthesis of MAX-DOAS validation campaigns in the IGP, TROPOMI structural biases, and logarithmic radical loss ratio constraints, the following FNR thresholds are highly recommended for operational use over Delhi and surrounding Indian megacities.

Photochemical Regime	FNR Threshold (Column)	Physical Chemistry Characteristics in Delhi
Strongly VOC-Limited	< 1.8	NO_x -saturated environments where OH radicals are primarily terminated by NO_2 . Ozone production is constrained by the availability of HO_x / RO_x radicals. Found predominantly in the dense urban core of Delhi (e.g., Wazirpur, Ashok Vihar)

		during the winter inversion (December-February) ⁴ .
Transitional (Mixed)	1.8 – 3.2	Peak ozone production efficiency. Ozone formation is highly sensitive to changes in both VOCs and NO_x . Often observed in the suburban fringes of the NCR (e.g., Faridabad, Noida), during post-monsoon transitional months (October-November), and during peak afternoon hours in the dry season ¹ .
Strongly NO_x-Limited	> 3.2	Radical abundant environments where the production of organic peroxy radicals outpaces NO_x emissions. Ozone production scales almost linearly with available NO_x . Common in rural downwind regions of the IGP and across the broader Delhi region during the summer monsoon (June-September) due to immense biogenic VOC fluxes ⁵ .

3.3 Sensor Physics and Threshold Dependency

It is vital for modelers to understand that an FNR threshold is not a universal physical constant; it is an observational metric heavily dependent on the vertical sensitivity of the instrument

capturing the data. The distribution of trace gases in the troposphere is not uniform. NO_2 in Delhi is heavily concentrated in the lowest 500 meters of the boundary layer due to primary combustion sources. In contrast, HCHO has both primary surface sources and secondary photochemical sources that can occur higher in the boundary layer as VOCs undergo oxidation

while being vertically mixed².

Ground-based MAX-DOAS instruments view the atmosphere from the surface upwards and are exquisitely sensitive to the lowest boundary layer where ozone production actually occurs.

Satellite sensors like OMI and TROPOMI are nadir-viewing, integrating the entire tropospheric column. Consequently, satellite thresholds inherently differ from surface-based thresholds. If a modeler attempts to utilize an FNR threshold derived strictly from surface monitors or

MAX-DOAS (which tends to see relatively more NO_2 compared to the bulk column), they will systematically misclassify the regime if applying that threshold to a satellite column dataset².

The 1.8 to 3.2 transitional bounds provided above are specifically calibrated for integrated tropospheric column analyses utilizing modern high-resolution satellite retrievals.

4. TROPOMI / Sentinel-5P Product Specifications

To operationalize the FNR for Delhi, data from the Sentinel-5 Precursor (S5P) TROPospheric Monitoring Instrument (TROPOMI) constitutes the global standard. TROPOMI offers unprecedented native spatial resolution, enabling the distinction of localized urban cores from suburban fringes—a requirement for accurate, city-scale regime mapping that previous sensors like OMI could not achieve.

4.1 Satellite Product Definitions and Versions

The exact products required for generating the FNR are the offline (OFFL) Level 2 or Level 3 tropospheric column densities.

- **Nitrogen Dioxide:** COPERNICUS/S5P/OFFL/L3_NO2. The specific data band required is the tropospheric_NO2_column_number_density¹⁶.
- **Formaldehyde:** COPERNICUS/S5P/OFFL/L3_HCHO. The specific data band required is the tropospheric_HCHO_column_number_density¹⁹.

TROPOMI features a native spatial resolution of 5.5 km × 3.5 km at nadir (upgraded from 7 km × 3.5 km in August 2019)⁸. The satellite operates in a sun-synchronous low Earth orbit, providing a daily revisit time with an overpass at approximately 13:30 local solar time²². This early-afternoon overpass is highly advantageous for photochemical studies, as it aligns closely with the diurnal peak of solar actinic flux, maximum boundary layer height, and peak ozone production.

The algorithmic versions of these products have evolved. Early studies relied on v1.2, while current operational analyses utilize v2.x processors. Furthermore, specialized scientific

retrievals, such as the Peking University OMI NO_2 (POMINO) algorithm, apply explicit corrections for aerosol optical effects and utilize high-resolution daily a priori profiles from GEOS-CF. Comparisons between POMINO and standard operational reprocessed products

over polluted regions demonstrate that standard operational tropospheric HCHO and NO_2 columns can be 10% to 20% lower than POMINO outputs due to different treatments of aerosol scattering and surface reflectance²³. For standard forecasting and modeling purposes, the operational Copernicus OFFL products remain the most robust and accessible choice.

4.2 Quality Assurance (QA) Mechanics

Rigorous Quality Assurance (QA) filtering is paramount when utilizing these products over the Indian subcontinent, particularly given the severe aerosol loadings and monsoonal cloud cover. The data providers encode a continuous qa_value variable ranging from 0.0 (error) to 1.0 (perfect retrieval) alongside every pixel to automate masking.

Trace Gas	Recommended qa_value Threshold	Masking Justification
NO_2	> 0.75	Strictly removes scenes with a Cloud Radiance Fraction (CRF) > 0.5, snow/ice covered surfaces, and retrievals with problematic Air Mass Factors (AMFs) or a priori profile errors. This strict threshold is highly necessary over the IGP to prevent cloud albedo from artificially inflating the retrieval ¹⁷ .
HCHO	> 0.5	HCHO retrievals are inherently noisier due to weaker absorption signatures in the UV spectrum. Relaxing the QA threshold to > 0.5 retains sufficient data density for spatial averaging while still removing heavily clouded pixels and failed spectral fits ¹⁷ .

5. Retrieval Biases Over Polluted Urban India

When deploying TROPOMI data over Delhi, the mathematical computation of the FNR must contextualize established retrieval biases. Validation campaigns utilizing ground-based MAX-DOAS over Delhi and nearby Mohali consistently indicate that TROPOMI systematically underestimates tropospheric NO_2 by 30% to 50% in highly polluted Indian environments, while occasionally overestimating HCHO by up to 30%².

This severe low bias in NO_2 is driven by three intertwined radiative transfer mechanisms specific to the TROPOMI retrieval algorithm and the unique environment of the IGP:

1. **A Priori Profile Dilution:** The TROPOMI retrieval algorithm calculates the Air Mass Factor (AMF)—the critical scalar used to convert the observed slant column density into a vertical column density—by utilizing vertical a priori NO_2 profiles generated by the TM5-MP chemical transport model²². Because TM5-MP operates at a very coarse $1^\circ \times 1^\circ$ spatial resolution, it artificially smears the intense, highly localized boundary-layer NO_2 spikes characteristic of Delhi over a vast grid box. This underrepresents the fraction of NO_2 located near the surface. Because satellite sensitivity degrades toward the surface, underestimating the surface concentration in the a priori profile directly skews the scattering weights, causing a severe underestimation of the true vertical column².
2. **Aerosol Optical Depth (AOD) Shielding:** The Indo-Gangetic Plain sustains some of the highest aerosol loadings on Earth. High concentrations of scattering and absorbing aerosols (especially secondary inorganic aerosols and black carbon from agricultural burning) elevate the effective scattering layer higher into the troposphere. The satellite effectively "sees" the aerosol layer and misses the NO_2 hidden beneath it in the lowest hundreds of meters of the boundary layer, leading to significant structural uncertainties. Research indicates that secondary inorganic aerosols can dictate up to 43.5% of the NO_2 retrieval uncertainty in urban agglomerations².
3. **Surface Albedo Constraints:** TROPOMI algorithms traditionally relied on coarse OMI-derived climatologies for surface reflectivity. In urban areas with rapidly changing infrastructure and complex land use like Delhi, incorrect albedo estimations degrade the AMF, further contributing to the negative bias²².

Conversely, the HCHO product suffers less from systematic low bias but exhibits extraordinary random noise. The standard error for a single-pixel daily TROPOMI HCHO retrieval is massive.

Statistical analyses show that daily HCHO column observations must exceed 1.2 to 1.5×10^{16} molecules cm^{-2} to attain a barely acceptable 20% to 30% standard error in the resulting FNR⁴. Over Delhi, while summer HCHO can reach these levels due to biogenic activity, winter values frequently fall below this noise floor, rendering daily pixel-by-pixel FNR computation statistically invalid.

6. Pre-Gridded Analysis-Ready Data Architectures

A critical bottleneck in integrating satellite atmospheric chemistry into operational numerical models is the necessity to process massive, ungridded Level 2 swath files stored in hierarchical NetCDF formats. For an ozone modeler seeking to integrate a regime map without developing a custom geospatial processing pipeline for orbital geometries, utilizing cloud-hosted, pre-gridded Level 3 products is essential.

Google Earth Engine (GEE) acts as the optimal platform for this requirement. GEE natively

hosts the COPERNICUS/S5P/OFFL/L3_NO2 and COPERNICUS/S5P/OFFL/L3_HCHO collections. The GEE ingestion architecture continuously pulls the raw Level 2 orbits from the Copernicus hubs and grids them into an analysis-ready planetary scale format, preserving the native spatial resolution¹⁶.

It is important to note that there is no native, pre-computed "FNR product" available as a single asset on GEE, NASA GES DISC, or the Copernicus Open Access Hub. The FNR is a derived product and must be calculated dynamically. However, GEE allows for the immediate, on-the-fly

mathematical division of the HCHO and NO_2 Image Collections natively in the cloud¹⁶. By defining the bounding box over the Delhi NCR, applying the strict QA masks, and mapping a `.divide()` operator across temporally matching images, a modeler can export a ready-to-use GeoTIFF of the FNR regime without ever downloading a single NetCDF file¹⁶.

7. Daily Dynamics vs. Seasonal Climatologies: A Cost-Benefit Analysis

A central architectural decision in rendering an aerosol-photolysis ozone model "regime-aware" is determining the temporal cadence of the FNR map. Should the model ingest a daily calculated FNR generated from the previous day's overpass, or does a coarse seasonal FNR climatology capture the requisite value?

The scientific consensus heavily favors the deployment of a seasonal climatology, as it is computationally cheaper, vastly more stable, and yields essentially the same empirical utility for a photolysis-constrained model.

Computing a per-day FNR over Delhi introduces fatal mathematical noise. Because the satellite column retrieval error for HCHO is so large, it accounts for 40% to 90% of the total relative error in the resulting FNR. Attempting to use a daily FNR map will result in a highly speckled, erratic

regime field that jumps between VOC-limited and NO_x -limited states arbitrarily due to retrieval noise⁴. Furthermore, during the monsoon season, persistent cloud cover over the IGP triggers the `qa_value` filters, resulting in vast swaths of missing data on any given day¹³. A daily forecast model attempting to ingest this data would face unpredictable grid-cell dropouts, rendering spatial boundary condition enforcement unstable.

By contrast, a seasonal climatology—averaging all valid pixels over a multi-year baseline for a specific season (e.g., all winter days from 2021 through 2023)—effectively drives the random retrieval noise of the HCHO sensor toward zero via the Law of Large Numbers⁴. The photochemical regime of a city is dictated by structural, slowly varying emission factors: the spatial layout of dense traffic corridors, the permanent location of industrial zones in the NCR, and the reliable seasonal cycles of temperature and biogenic emissions. These structural realities do not flip wildly from day to day; they shift predictably on a seasonal basis².

A seasonal FNR climatology over Delhi buys the forecast immense structural stability. It guarantees that the highly polluted urban core is consistently tagged as VOC-limited during the

winter inversion, and the downwind rural fringes are tagged as NO_x -limited during the summer biogenic peak. For a model seeking broad regime-awareness without absorbing daily

computational overhead or dealing with data dropouts, the climatological approach captures the vast majority of the actionable atmospheric physics at a fraction of the processing cost.

8. Concrete Implementation Recipe for Delhi

To integrate the satellite-derived FNR into the MCM v3.3.1 clear-sky model workflow, the following precise protocol must be executed within the Google Earth Engine JavaScript API or Python API.

Phase 1: Data Sourcing and Temporal Definition

1. **Data Source:** Google Earth Engine (ee.ImageCollection).
2. **Target Collections:** COPERNICUS/S5P/OFFL/L3_NO2 and COPERNICUS/S5P/OFFL/L3_HCHO¹⁶.
3. **Cadence:** Generate a Seasonal Climatology utilizing a 3-to-4-year baseline (e.g., 2021–2024) to eliminate interannual meteorological anomalies and the extreme emission perturbations of the 2020 COVID-19 lockdowns¹⁰.
 - *Winter (DJF)*: December, January, February.
 - *Pre-Monsoon (MAM)*: March, April, May.
 - *Monsoon (JJA)*: June, July, August.
 - *Post-Monsoon (SON)*: September, October, November.

Phase 2: Processing Steps

1. **QA Masking:** Apply an embedded function mapping over both collections to mask low-quality pixels. For NO_2 , drop pixels where $qa_value < 0.75$ ¹⁷. For HCHO, drop pixels where $qa_value < 0.5$ ¹⁷.
2. **Temporal Reduction:** Reduce both masked image collections utilizing a `.mean()` or `.median()` reducer to create a single static image of NO_2 and a single static image of HCHO for the target season¹⁶.
3. **Unit Harmonization:** Both GEE OFFL products are natively provided in mol/m^2 . Verify that no arbitrary scaling factors have been accidentally applied.
4. **Ratio Computation:** Use the `.divide()` operator to compute the ratio: `HCHO_image.divide(NO2_image)`³².
5. **Spatial Smoothing (Optional but Recommended):** Apply a mild focal mean (e.g., a 3x3 pixel moving window) to the resulting FNR image to smooth boundary discontinuities between the 5.5 km grid cells. This prevents numerical shocks when the FNR map is ingested into the chemical transport model.
6. **Export:** Export the final GeoTIFF clipped specifically to the Delhi NCR bounding box (e.g., 28.2°N to 29.0°N, 76.8°E to 77.5°E)¹⁶.

Phase 3: Regime Classification Mapping

Ingest the exported GeoTIFF into the modeling framework and apply the threshold classification to every spatial grid cell based on the polluted Indian megacity standards¹:

Grid Cell FNR Value	Assigned Regime Label	Empirical Model Action
FNR < 1.8	Regime 1 (VOC-Limited)	Apply scaling to mimic titration and OH suppression.
$1.8 \leq \text{FNR} \leq 3.2$	Regime 2 (Transitional)	Peak baseline PSS efficiency; apply minimal dampening.
FNR > 3.2	Regime 3 (NO_x -Limited)	Apply positive scalar to emulate radical-driven net O_3 production.

9. The Limitations of FNR in a VOC-Absent Modeling Environment

While generating and mapping this FNR climatology provides a visually compelling data layer, it is critical to provide an honest, chemically rigorous assessment of what this regime label *can* and *cannot* improve given the highly specific constraints of the user's photochemical model. The stated model framework restricts itself to the aerosol-photolysis pathway utilizing MCM v3.3.1 clear-sky $J(NO_2)$ and $J(O^1D)$, entirely lacking VOC measurements and explicit VOC degradation mechanisms.

9.1 The Fundamental Constraint of the Photostationary State

As outlined in Section 2, the photolysis of NO_2 and subsequent reaction of $O(^3P)$ with O_2 creates ozone, but the reaction of ozone with NO destroys it just as quickly. In a purely inorganic atmosphere, this creates a rapid null cycle where $[O_3]$ is strictly determined by the ratio of J_{NO_2} to k_3 . There is no mechanism for net ozone accumulation.

The fundamental purpose of an FNR regime map is to estimate the abundance of RO_2 and HO_2 radicals relative to NO_x . These radicals are the only chemical agents capable of converting NO to NO_2 without destroying an ozone molecule in the process, thus breaking the photostationary null cycle and allowing ozone to accumulate.

If the numerical model lacks VOC measurements entirely, it possesses no chemical feedstock to generate RO_2 or HO_2 . It is physically restricted to simulating the Leighton null cycle. Therefore, the numerical solver cannot organically compute net ozone production regardless of what the FNR map indicates.

9.2 What an FNR Regime Label CANNOT Do

- **It cannot mechanistically synthesize missing VOCs.** The solver requires discrete initial concentrations of species like isoprene, toluene, or formaldehyde to trigger the MCM v3.3.1 oxidation chains. An FNR ratio is a dimensionless number; it cannot substitute for chemical boundary conditions.
- **It cannot spin up a radical pool.** Without explicit degradation reactions, the model cannot calculate the concentration of OH , HO_2 , or RO_2 , rendering the true chemical production rate equations unsolvable.
- **It cannot capture complex secondary aerosol formation.** The degradation of VOCs leads to secondary organic aerosols (SOA), which subsequently alter the AOD and feedback into the photolysis rates. The FNR label cannot emulate this physical feedback loop.

9.3 What an FNR Regime Label CAN Do: Empirical Scaling

If the model lacks VOC chemistry, the only mathematically viable way to utilize the FNR map is as an empirical, statistical scaling factor—essentially acting as a spatial parameterization of the Ozone Production Efficiency (OPE).

By overlaying the seasonal FNR climatology onto the spatial grid, the modeler can artificially force the output to deviate from the PSS equation. The implementation functions as a spatial mask:

- In **Regime 1 (VOC-limited)** grid cells ($FNR < 1.8$), the real atmosphere is saturated with NO_x , which severely suppresses the OH radical and titrates ozone. The model algorithm can be programmed to artificially depress the final O_3 concentrations output by the PSS calculation to emulate this nocturnal and wintertime titration effect.
- In **Regime 3 (NO_x -limited)** grid cells ($FNR > 3.2$), massive unmeasured biogenic VOCs are driving the production of RO_2 and HO_2 . The model can apply an empirically derived multiplication factor to the PSS output—scaling the theoretical O_3 up by a factor proportional to the ambient NO_2 concentration and the unattenuated $J(O^1D)$ —to mimic radical-driven net production.

Ultimately, integrating a seasonal FNR climatology into a VOC-absent model transforms a purely theoretical photostationary calculator into an observationally constrained, regime-aware empirical model. It acts as a highly effective, computationally inexpensive spatial patch, forcing

the model to emulate the end results of the complex VOC- HO_x - NO_x cycle across the heterogeneous landscape of Delhi without incurring the massive computational penalty of simulating the organic chemistry itself.

10. Conclusion

The integration of Sentinel-5P TROPOMI HCHO and NO_2 tropospheric column data offers a powerful, observation-based mechanism to evaluate ozone production sensitivities over the extreme photochemical environment of the Indo-Gangetic Plain. Traditional global FNR thresholds derived from North American environments must be discarded in favor of localized

parameters for Indian megacities. Due to intense baseline NO_x pollution, rapid boundary layer dynamics, and statistical retrieval uncertainties, the transitional regime for Delhi sits broadly between an FNR of 1.8 and 3.2.

Attempting to resolve daily FNR maps is counterproductive due to the severe signal-to-noise ratio of the HCHO sensor—which requires massive column densities to achieve acceptable standard errors—and extensive cloud masking during the Indian summer monsoon. Instead, leveraging cloud-based platforms like Google Earth Engine to process Level 3 offline data into stable, multi-year seasonal climatologies provides an operationally superior and cost-effective dataset. This climatological approach captures the structural regime variations, such as the

wintertime VOC-limited urban core and the summertime NO_x -limited rural fringes, while smoothing out retrieval noise.

However, mathematical proxies cannot defy the laws of physical chemistry. If the underlying photochemical model utilizes the Master Chemical Mechanism restricted solely to aerosol-attenuated photolysis pathways and entirely lacks VOC input data, the kinetic solver can only resolve a zero-net photostationary state. The FNR regime map cannot inject missing chemical kinetics or spin up a radical pool. Its true utility in this highly specific modeling environment is to serve as an intelligent spatial mask, allowing the modeler to apply regime-dependent empirical scaling factors. By doing so, the modeler can mathematically mimic the non-linear realities of Delhi's ground-level ozone production, bridging the gap between an inorganic photolysis calculator and the true complexity of urban air pollution.

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